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RESEARCH ARTICLE

ANTIMICROBIAL EFFECT OF LIPOXYGENASE MEDIATED PRIMARY PHYTO-OXYLIPINS FROM HORSE GRAM (*Dolichos biflorus*) GERMINATING SEEDLINGS ON COMMON MICROBIAL FLORA

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ABSTRACT

Plant oxidized fatty acids, phyto-oxylipins, have had a great impact on plant physiology. Phyto-oxylipins are a group of biologically active molecules that play a decisive role in plant defense against plant pathogens (Feussner and Wasternack, 2002;). Purification and characterization of the metabolites by plant Lipoxygenases were well studied when compared to the physiological role of Lipoxygenases. We doubted the role of Phyto - oxylipins in providing resistance to the Horse gram (*Dolichos biflorus*) plant against the common plant pathogens and play a key role in plant defense responses. Hence we made an attempt to find out the antimicrobial activity of primary phyto-oxylipins generated from *Dolichos biflorus* Lipoxygenase against some of the common bacterial and fungal strains in vitro.

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INTRODUCTION

Lipoxygenases (LOX) produces the Phyto-oxylipins by the oxygenation of polyunsaturated fatty acids to form 9- or 13-hydroperoxides (Mita *et al.*, 2007). Many of the physiological events in plant cells such as plant defense, development, and senescence are believed to be governed by the Phyto-oxylipins (Liu and Han, 2010). LOXs catalyses the hydroperoxidation of Poly unsaturated fatty acids (PUFAs) containing a cis, cis-1, 4 pentadiene moiety, such as linoleic acid (LA) (C18:2) or linolenic acid (C18:3) to form Hydroperoxyoctadecatrienoic acid (HPOTs) (sabineRoshal *et al.*, 2001). It is the primary step of the Lipoxygenase pathway, a pathway from which most of the Phyto-oxylipins were identified and described. The newly formed Lipid hydroperoxides have a conjugated double bond system which forms absorption chromophore at 234 nm in an UV spectrophotometer (James N. Siedow: 1991). Only 9S- or 13S-hydroperoxides were generated when Lipoxygenases acted up on their substrates (PUFAs). It is due to the fact that the LOX enzyme is stereo-specific, and hence plant LOXs are commonly referred as 9- or 13-LOX (Feussner and Wasternack, 2002, Porta and Rocha-Sosa, 2002). During the utilization of Linoleic acid, only 13- Hydroperoxy products were produced by Type 1 lipoxygenases where as Type 2 lipoxygenases produce both 9-, 13- Hydroperoxide products in a 50:50 ratio (James N. Siedow: 1991). These LOX Hydroperoxides (9- and 13-hydroperoxides) have been shown a profound effect on viability of plant cell and also localized cell death regulated in the hypersensitive reaction (Rusterucci *et al.*, 1999). The 13-LOX derived compounds, including 13-octadecadienoic acid (13-HPODE) and 13-hydroxyoctadecatrienoic acid (13-HPOTrE) are regulators of plant defense gene expression (Almeras *et al.*, 2003). However, the defense related functions of 9-LOX are obscure.

Horse gram (*Dolichos biflorus*) is a leguminous crop and is largely grown in hot semiarid areas of Southern India. This plant is highly disease resistant against the pathogenic micro flora. There were only few fungal diseases like root rot, anthracnose, rust and viral infections like Yellow mosaic disease to mention which infect the horse gram plant. This plant shows great resistance towards the attack by normal microbial flora, which interferes with the physiological functions of the plant. Hence we selected this plant to check the anti-microbial properties of the phyto-oxylipins isolated from the partially purified horse gram Lipoxygenase against some of the common microbial flora. The formation of biologically active molecules from fatty acid 9-hydroperoxides in plants has not received great attention, but biological role for such compounds in plant defense has been suggested in potato and tobacco in response to pathogen attack as a result of formation of colneleic acid (CA) and colnelenic acid (CnA) (Weber *et al.*, 1999). The divinyl ethers (CA and CnA) of the 9-LOX, stomatal defense response against microbial invasion to plant leaf (Montillet *et al.*, 2013).

MATERIALS AND METHODS

Growth of Horse Gram Seedlings

The horse gram seeds (var.HPK-2) obtained from local farmlands in Anantapur district, Andhra Pradesh; India, were surface sterilized and the seeds soaked in distilled water were grown as seedlings. The horse gram germinated seedlings were harvested at 72 hours interval from the day of germination and used for extraction of Lipoxygenase enzyme

Determination of Lox Activity

LOX activity in Horse gram seedlings was determined spectrophotometrically by monitoring the formation of

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conjugate diene at 234 nm. Seedling samples were prepared by homogenizing the harvested seedlings in 100 mM potassium phosphate buffer (pH 7.0). The homogenates were centrifuged at 10,000 rpm for 30 min at 4°C and the resulting supernatants were used for LOX activity determination in 2.9 ml of 100 mM potassium phosphate buffer (pH 7.0). The reaction was initiated by the addition of 10 μ l of 100 μ M Linoleic acid as substrate and the formation of conjugate diene at 235 nm was monitored with a Shimadzu 160 UV-VIS spectrophotometer for 3 min. The LOX activity was calculated using a ϵ of 27,500 M⁻¹cm⁻¹. One unit of Lipoxygenase activity is defined as 1 μ mole of hydroperoxides formed per minute. Specific activity is expressed as units per mg protein. LOX activity measurements were performed thrice with seedling material collected at every time point in three independent experiments.

Preparation of LOX hydroperoxides

Partially purified Horse gram Lipoxygenase was extracted as per the method described by Reddanna *et al.*, 1990 and used for the preparation of the hydroperoxides. The reaction was initiated with linoleic acid (200 μ M) under constant shaking at 37 °C. The reaction was ended by acidifying the reaction mixture to pH 2.5 with 6 N HCl and the products formed were extracted into two volumes of hexane: ether (1:1) twice. The organic solvent removed from the aqueous layer in a separating funnel and passed through anhydrous granular sodium sulphate. Rotator evaporator was used to evaporate the organic phase to total dryness. After re-dissolving, the dried products were analysed through UV SPEC and HPLC.

Antimicrobial activity of LOX metabolites-filter disc method

Antibacterial activity of hydroperoxide products of horse gram LOX was assayed against the test organisms (*Pseudomonas aeruginosa*, *Bacillus subtilis*, *Escherichia coli* and *Proteus vulgaris*). The bacterial strains were inoculated into nutrient broth and incubated at 37°C for 24 h. 1 ml of the culture was spread onto the previously prepared nutrient agar plates. Hydroperoxides were weighed down onto the filter discs and then incubated for 24 h at 37°C and studied for the antimicrobial property by measuring the inhibition zone. We repeated the same experiment to check the antimicrobial activity reproducibility after 1 week of the LOX hydroperoxides preparation and ended in similar findings (fig 2). Vancomycin (30mg) was used as control antibiotic in this experiment. Similarly, Antifungal activity of Hydroperoxides of LA against the test fungi (*Aspergillus niger*, *Rhizopus sorysae*, *Trichoderma harzianum* and *penicillium notatum*) was assayed by filter disc assay method and also by agar diffusion method. The above mentioned fungi were grown on potato dextrose broth on a reciprocal shaker for four days at 25°C to obtain the conidial suspensions. About 1 ml of spore suspension (2.5×10^4) was pipette onto potato dextrose agar plates and spread evenly with a glass spreader. To test the effect of hydroperoxides on the growth of above mentioned organisms on potato dextrose agar, sterile 14 x 2 mm filter discs (Whatman No.1), were placed on the agar surface of previously seeded potato dextrose agar plates. LOX hydroperoxides dissolved in 10 μ l of 10% methanol were pipetted on to the filter discs on petriplates. Plates were incubated at 25°C for 72 hours and examined for zones of growth inhibition around each disc. The lowest concentration of LOX hydroperoxide that produced a detectable zone of inhibition was considered as minimum inhibitory concentration (MIC). The experiment was repeated 3 times. In agar diffusion method, a small well in proportionate to the size of the disc was created on the plate and is

loaded with minimum inhibitory concentration of LOX hydroperoxides and examined for the inhibitory zone. Dissimilar results were obtained when we checked for the reproducibility of anti fungal activity after 1 week.

RESULTS AND DISCUSSION

Antibacterial activity studies by filter disc Assay method showed that the LOX Hydroperoxides possess better antibacterial activity compared to the commercially available antibiotics like Gentamycin and Kanamycin. A minute concentration of 2 μ l (1mg/ μ l) of LOX hydroperoxides produced inhibitory zones of 14, 26, 30, and 28 mm diameters in *Pseudomonas aeruginosa*, *Escherichia coli*, *Bacillus subtilis* and *Proteus vulgaris* cultured plates respectively. Whereas, a 30 mg concentration of Gentamycin formed inhibitory zones with diameters 30, 40, 44 and 42 mm which are comparatively very lower than LOX hydroperoxides. The results of antibacterial activity were tabulated in table 1 and shown in figure 1 and 2.

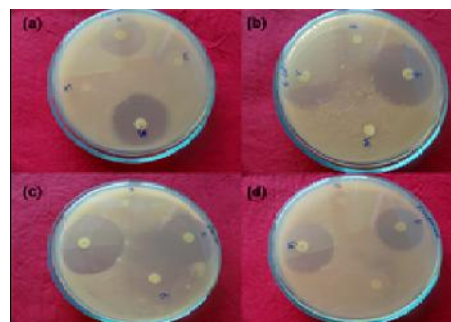


Figure 1 Antibacterial activity of Horse gram Lipoxygenase mediated Lipid Hydroperoxides on cultured species of a) *Pseudomonas aeruginosa* b) *Bacillus subtilis* c) *Escherichia coli* and d) *Proteus vulgaris* when compared with Gentamycin (G-30mg Disc) and Kanamycin (K-30mg Disc). E represents lipid hydroperoxide extract – 2mg and M represents Methanol – 2 μ l

The effect of Horse gram LOX hydroperoxides on the growth of various fungal species *Aspergillus niger*, *Rhizopus sorysae*, *Trichoderma harzianum*, and *penicillium notatum* were also tested by both filter disc and agar diffusion methods and the results were tabulated (Table 2) (Figure 3). Antimicrobial activity Reproducibility of LOX hydroperoxides were positive against the bacterial strains and null against the fungal strains after 1 week of their preparation.

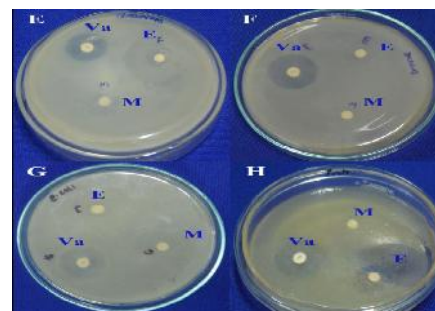


Figure 2 Antibacterial activity of Horse gram Lipoxygenase mediated Lipid Hydroperoxides on cultured species of e) *Pseudomonas aeruginosa* f) *Bacillus subtilis* g) *Escherichia coli* and h) *Proteus vulgaris* when compared with Vancomycin (G-30mg Disc), to check their activity after one week. E represents lipid hydroperoxide extract – 2mg and M represents Methanol – 2 μ l

We are well aware that Most of the Investigations in plant pathogen resistance were restricted mainly to Jasmonic acid biosynthesis in plant defense-signaling pathways. There are reports regarding the essentiality of oxylipins other than

jasmonates in plant protection against the pathogens (Blee, 2002), Conidial germination of rice blast in *Pyricularia oryzae* was inhibited by LOX hydroxides (13-HOTrE and 9-HOTrE) (Shimura *et al.*, 1983) and in *A. niger* by LOX hydroperoxides (13-HPODE and 13-HPOTrE) (Sailaja *et al.*, 1998). Inhibition of cyst spore germination of *Phytophthora capsici* by 9- and 13-HPOTrEs and hydroxyl derivatives of arachidonic acid were reported (Ricker and Bostock, 1994).

pathway, 2-Hexenal compounds, produced by the action of Hydroperoxide Lyases of 13 HPOTs shows a moderate antimicrobial activity (Kevan P. C. Croft; 1993). No reports were available on the antimicrobial activity of the Ketofattyacids produced during this reaction, having absorption maxima at 270nm (Peak 2 in Fig 4).

Table 1 Minimum inhibitory concentration (MIC) of the Horse gram Lipoxigenase mediated Lipid hydroperoxides against various bacterial species-disc diffusion Method

Organism	Diameter of the clear zones formed by minimum Inhibitory concentration of LOX Hydroperoxides (2 µg/disc)	Diameter of the clear zones formed by Control (Methanol)	Diameter of the clear zones formed by Gentamycin (30 µg/disc)	Diameter of the clear zones formed by Kanamycin (30 µg/disc)
<i>Pseudomonas aeruginosa</i> (a)	14 mm	2 mm	30 mm	28 mm
<i>E.coli</i> (b)	26 mm	4 mm	40 mm	34 mm
<i>Bacillus subtilis</i> (c)	30 mm	16 mm	44 mm	40 mm
<i>Proteus vulgaris</i> (d)	28 mm	3 mm	42 mm	36 mm

Table 2 Minimum inhibitory concentration (MIC) of the Horse gram Lipoxigenase mediated Lipid hydroperoxides against various fungal species-disc diffusion Method and Agar diffusion method

Organism	Diameter of the clear zones formed by Disc diffusion method of LOX Hydroperoxides (2 µg/disc)	Diameter of the clear zones formed by Agar diffusion method of LOX Hydroperoxides (2 µg/disc)
<i>Aspergillus niger</i>	14 mm	15 mm
<i>Rhizopus oryzae</i>	8 mm	11 mm
<i>Trichoderma harzianum</i>	9 mm	13 mm
<i>Penicillium notatum</i>	9 mm	12 mm

Thus the LOX metabolites generated in the LOX pathway act as antimicrobial compounds during plant defense responses against pathogen infection and helps in the resistance development. These fatty acid hydroperoxides are extremely reactive causing oxidative membrane damage which in turn leads to cell necrosis and death (Hildebrand, 1989). In our present study, horse gram LOX products of LA were used to test such possibility and their anti-microbial effects were screened by filter disc assay and agar diffusion method. These assays showed that hydroperoxides of LA were effective against *Aspergillus niger*, *Rhizopus oryzae*, *Trichoderma harzianum*, and *penicillium notatum* as antifungal agents.

Table 3 UV absorption spectra of Horse gram Lipoxigenase mediated lipid hydroperoxides

No	Peak or valley	Wave length (nm)	Absorbance
1	Peak	283.00	0.156
2	Peak	208.00	0.215
1	Valley	247.00	0.078
2	Valley	199.00	0.083

Similarly These LA hydroperoxides were found to be more effective antimicrobial agents than the commercially available antibiotics against the bacterial species, when we tested them by filter disc method on some bacterial cultures of *Pseudomonas aeruginosa*, *Escherichia coli*, *Bacillus subtilis* and *Proteus vulgaris*. The present study indicates that the 9-LOX and 13-LOX metabolism are involved in generation of anti-microbial oxylipins in Horse gram against various fungal and bacterial species as these phyto-oxylipins having UltraViolet absorption maxima at 234nm (James N. Siedow; 1991), which we can clearly observe in peak 1 in the UV chromophore (Fig 4).

Though most of the Phyto-oxylipins of 13-LOX involved in synthesizing signaling compounds in the Lipoxigenase

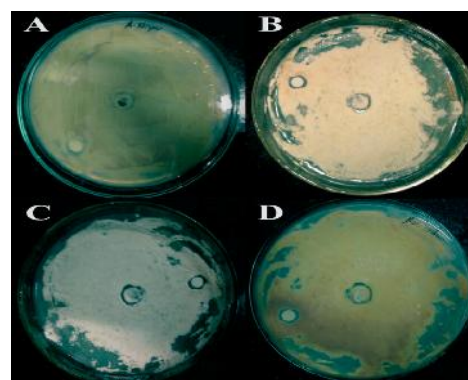


Figure 3 Antifungal activity of Horse gram Lipoxigenase mediated Lipid Hydroperoxides on cultured species of a) *Aspergillus niger*, b) *Rhizopus oryzae*, c) *Trichoderma harzianum* and d) *Penicillium notatum*. E represents lipid hydroperoxides extract – 2mg

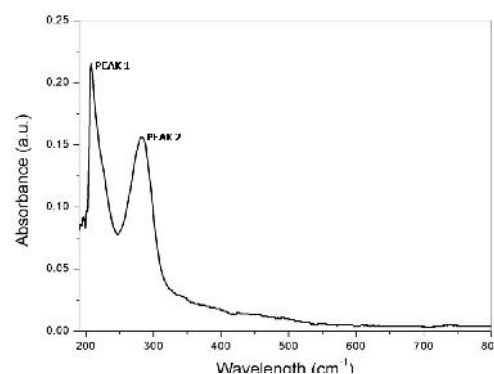


Figure 4 UV spectrogram showing peaks representing Horse gram Lipoxigenase mediated primary lipid Hydroperoxides. Peak 1 ($\lambda_{\max}$ 234 nm) refers to the Lipid Hydroperoxide products (9-, 13- primary phyto oxylipins). Peak 2 refers to the ketofattyacids ($\lambda_{\max}$ at 270 nm)

CONCLUSION

The primary products of LOX, the fatty acid hydroperoxides, are very reactive and cause oxidative damage to membranes, leading to cell necrosis and death (Hildebrand, 1989). LOX metabolites in this study exerted antimicrobial activity against various fungal and bacterial pathogens. The LOX hydroperoxides formed from LA are potent antimicrobial agents and the LA pathway is operative in the pathogen infected conditions, suggesting their possible involvement in the development of resistance.

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